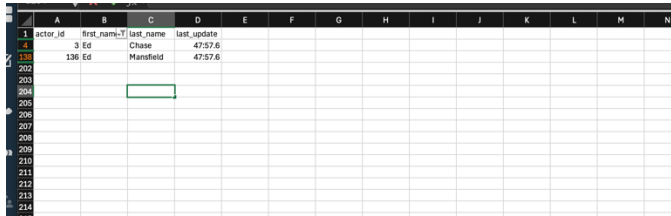


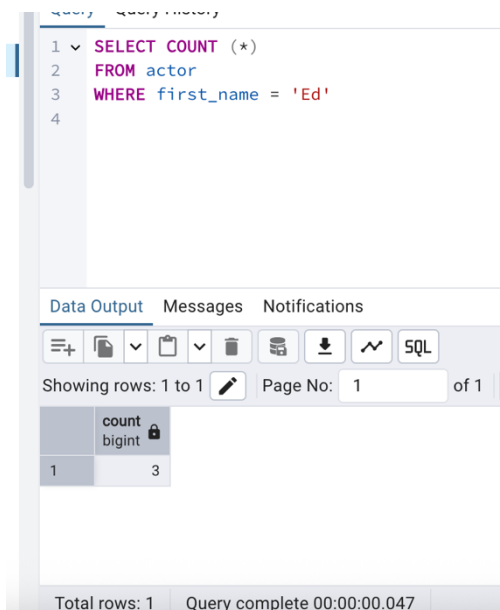
Exercise 3.1 Yaw Assensoh Opoku

2a. I used the filter option and sorted it out.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	actor_id	first_name	last_name	last_update										
2	3	Ed	Chase	47:57.6										
3	136	Ed	Mansfield	47:57.6										
4														
5														
6														
7														
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22														
23														
24														

b. Correct statement makes it easy and fast but can be a little frustrating the first time and the answers doesn't match as shown in the images inserted.



```
1 SELECT COUNT (*)
2 FROM actor
3 WHERE first_name = 'Ed'
4
```

	count
1	3

Total rows: 1 Query complete 00:00:00.047

c. I do find Excel much easier because I've been using throughout the past achievements and have grips on it which makes it easier to work with, so my biased answer will be Excel.

However, as mentioned earlier, with the right statement and correct input, SQL statemen is faster and should be easier than Excel and I'm up for the challenge.

3.a Execute the following query and list the names of the columns in the payment table.

```
SELECT * FROM payment LIMIT 10;
```

Payment_id, customer_id, staff_id, rental_id, amount,
payment_date

3b. Under the “table_name” column, what are the names of the tables that are available in the Rockbuster database? (List all names.)

```
SELECT * FROM information_schema.tables  
WHERE table_schema = 'public'  
AND table_type = 'BASE TABLE'
```

Actor, store, address, category, city, country, customer, film_actor,
film_category, inventory, language, rental, staff, payment, film

3c. Within the pgAdmin 4 console, can you think of another way to list all the table names in the database instead of the SQL statement above?

In pgAdmin 4, click on Rockbuster, then Schemas, then Public and then click on Tables and all the list of table names are going to show up.

3d. Analyze the rental duration distribution. How many days are most films rented for?

Most films are rented for 6 days which has 212 as number of films. Here's the graph to prove it.



4. Consider who in Rockbuster Stealth might want to use an OLAP or OLTP system for their data needs; for example, the sales department, which is interested in sales trends, would likely use an OLAP system. Describe at least two situations for each type of system.

- Sales Department: The sales team at Rockbuster Stealth would use OLAP to analyze sales trends over a period of time for inventory by analyzing trends by genre or region.
- Management Team: For the company's growth and decision making, the team would use OLAP to gain insights into the business performance. E.g. Comparing revenue from different genres of movies.
- Inventory management: would use OLTP to update and manage real time data on stocks and ensuring to put out what the customer needs.

- Operations Department: Would use OLTP to process real time rentals and payment .

5. Does the invoice contain structured or unstructured data? Write an explanation for your answer.

- The invoice is structured data because the information is well organized in a clear format.

Invoice_ID	Client_Full_Name	Client_Address	Item_Code	Quantity	Item_Description
2019100	Mr. Timothy Walker	40 Sheila La Sparks,NV	001	1	New Video Collection Licensing

Unit_Price	Subtotal	Company_Name	Company_Address	Payer_Name	Account_Number
\$730	\$730	Oaklanders	4826 Norma Avenue Anderson, TX	Miko Santo	4929 3310 0057 5422